

WINDOWS ENTERPRISE ADMINISTRATION & ACTIVE DIRECTORY SECURITY LAB

Consolidated Portfolio Report

Active Directory | DNS & DHCP | Group Policy | Forest Trusts | Delegated Administration | File
Services | PowerShell | Troubleshooting | Security Validation

AD DS	DNS / DHCP	GPO / ACLs	PowerShell / Security
-------	------------	------------	-----------------------

Prepared by Ayinde Perouza

Trinidad and Tobago | August 2026

Lab-only technical documentation. No production credentials or production data are included.

1

Executive Summary

What the project demonstrates and how the consolidated report is scoped

This report consolidates the Windows enterprise administration work completed across multiple hands-on lab exercises into one portfolio-ready case study. The core environment centers on Windows Server Active Directory, DNS, DHCP, Windows 11 clients, Group Policy, identity and access administration, file services, PowerShell and security-focused validation. A separate GOODCORP.LOCAL administration challenge is included where it demonstrates least-privilege delegation and Active Directory ACL engineering.

The project goes beyond successful configuration. It records failed states, troubleshooting decisions, remediation steps and validation evidence. That evidence-based approach is important in enterprise administration because a configuration is not complete until its effect is verified and its security impact is understood.

Portfolio positioning The report is intentionally broader than a forest-trust-only case study. It supports the resume title “Windows Enterprise Administration & Security Lab” by showing end-to-end administration, troubleshooting and secure access-control work.

Primary outcomes

- Built and administered a Windows Server Active Directory environment with AD DS, DNS, DHCP, users, groups, OUs, permissions and Windows clients.
- Implemented and validated a two-way transitive forest trust between TT.WI.LOCAL and BB.WI.LOCAL after resolving cross-forest DNS issues.
- Applied Group Policy for endpoint restrictions, remote logon controls and user configuration, then validated policy application on Windows clients.
- Implemented least-privilege Helpdesk delegation using object-specific Active Directory access-control entries rather than broad account-management rights.
- Worked with SMB/NTFS permissions, folder redirection and Volume Shadow Copy / Previous Versions for file-service administration and recovery.
- Used PowerShell and native Windows administration tools to diagnose identity, DNS, service-state, network, permissions and endpoint issues.
- Extended Windows visibility with Sysmon and later telemetry integration into a security monitoring lab.

Administration methodology

Step	Phase	Purpose
1	Configure	Implement the required service, policy, access rule or relationship.
2	Test	Verify expected behaviour from the correct user, client and network context.

Step	Phase	Purpose
3	Diagnose	Use commands, logs and policy/ACL inspection to isolate the failure.
4	Remediate	Correct the proven cause without broadening access unnecessarily.
5	Validate	Re-test functionality and security boundaries; preserve evidence.
6	Document	Record configuration decisions, troubleshooting results and operational lessons.

Scope boundaries

- Core topology: TT.WI.LOCAL and BB.WI.LOCAL Active Directory forests on the 10.1.10.0/24 VMware lab network.
- Separate challenge environment: GOODCORP.LOCAL, used for the Helpdesk least-privilege delegation exercise.
- Operational ticket exercises such as VSS recovery and OpenSSH remediation are included as administration case studies, not as permanent components of the core forest-trust topology.

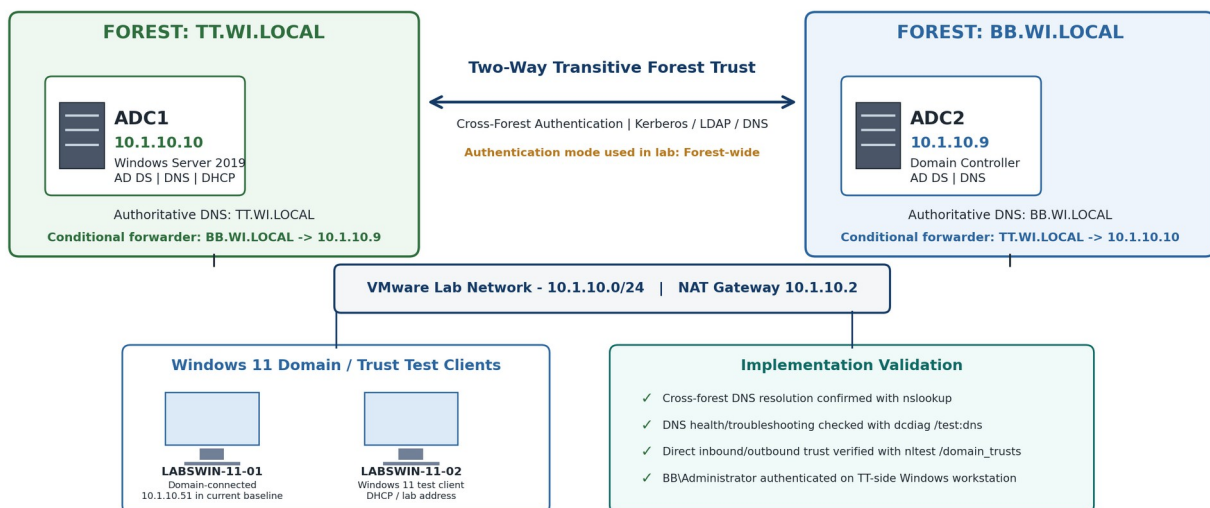
2 Architecture & Environment

Core lab topology, roles and administration boundaries

Core Active Directory topology

Windows Enterprise Administration - Core Active Directory Topology

TT.WI.LOCAL <-> BB.WI.LOCAL | DNS Conditional Forwarders | Two-Way Transitive Forest Trust



Scope note: The GOODCORP.LOCAL helpdesk-delegation challenge was completed in a separate logical lab and is not shown in this core topology.

All hostnames, domains and IP addresses shown are lab-only values used for portfolio documentation.

Figure 1. Verified core topology used for DNS, trust and authentication testing.

Core environment inventory

Asset	Domain / Context	Role	Key Services / Purpose	Address
ADC1	TT.WI.LOCAL	Windows Server 2019 domain controller	AD DS, DNS, DHCP	10.1.10.10
ADC2	BB.WI.LOCAL	Independent forest domain controller	AD DS, DNS	10.1.10.9
LABSWIN-11-01	TT-side client	Windows 11 administration / test workstation	Domain and authentication testing	10.1.10.51 in current baseline
LABSWIN-11-02	Lab client	Windows 11 trust / authentication test client	Cross-domain testing	DHCP / lab address
VMnet8	Lab network	VMware NAT/internal segment	Connectivity between lab systems	10.1.10.0/24; gateway 10.1.10.2

Why the architecture matters

The two forests were intentionally independent. That forced cross-forest authentication to depend on working DNS resolution, the forest trust and endpoint authorization policies rather than on a single-domain shortcut. This provided a realistic way to practice the dependency chain between name resolution, Kerberos/LDAP authentication and Windows logon policy.

Design principle Active Directory troubleshooting starts with dependencies. DNS must be reliable before a trust can be considered healthy, and successful authentication still does not guarantee that local or domain policy will authorize interactive logon.

3 Active Directory Domain Services & Identity Administration

Users, groups, OUs, Windows clients and policy structure

Administration capabilities practiced

- Domain controller deployment and administration with Windows Server.
- User and group creation, membership administration and access-control testing.
- Organizational Unit creation and use as a policy/delegation boundary.
- Windows workstation enrollment and domain-based authentication testing.
- Group Policy creation, linking, update and verification.
- Identity and privilege troubleshooting using whoami, PowerShell security-token checks and Active Directory cmdlets.
- Security documentation that records both configuration intent and the validation used to prove it.

Restricted-user / endpoint-policy exercise

A restricted domain-user scenario was used to practice how enterprise administrators provide the access a user needs while limiting unnecessary administrative capabilities. The user was placed within the UserAccounts policy scope, permitted to use the workstation remotely and subjected to user-configuration restrictions.

Control Area	Implementation	Administrative Rationale
Remote access	RDP permitted for the intended user/group	Business access remained available while logon rights were controlled.
Desktop / personalization	Wallpaper/theme-related settings enforced	Reduced user ability to alter managed endpoint configuration.
Administrative tools	Administrative utilities and selected control surfaces restricted	Limited access to tools not required for the user role.
Folder redirection	Documents redirected to a server-hosted location	Centralized user data and separated endpoint storage from user files.
Policy validation	gpupdate /force, gpresult /r and RSOP used during testing	Confirmed that policy was applied from the expected scope.

Group Policy operating practices

- Used dedicated GPOs for specific controls rather than treating the Default Domain Policy as a catch-all configuration location.
- Applied policy to the intended OU or security scope and then verified the resulting client state.
- Distinguished authentication from logon authorization; valid credentials can still be denied interactive logon by policy.
- Used gpupdate /force for immediate application during testing, then gpresult /r or RSOP for evidence of applied policy.

```
gpupdate /force
gpresult /r
rsop.msc
```

4

DNS, DHCP & Name-Resolution Engineering

Dependency troubleshooting before cross-forest trust configuration

Initial DNS symptoms

- nslookup timeouts and inconsistent server identification.
- “Server: Unknown” results associated with incomplete reverse lookup/PTR configuration.
- Cross-forest name-resolution failures between TT.WI.LOCAL and BB.WI.LOCAL.
- dcdiag Forwarders findings that required interpretation in the context of an isolated/internal lab.
- IPv6 loopback (::1) precedence on ADC2 contributing to confusing DNS delays.
- Recursion/root-hint behaviour introducing delays that were not required for the internal-only lab phase.

Remediation sequence

Area	Action	Validation Result
ADC1 local DNS	Confirmed AD-integrated records and service health; validated SRV records under _msdcs/_tcp including LDAP/Kerberos; created/validated PTR data for 10.1.10.10.	adc1.tt.wi.local resolved correctly and reverse lookup identified the server.
ADC2 local DNS	Corrected NIC DNS so 10.1.10.9 was used as the DNS server; addressed ::1 precedence; created/validated PTR data; checked DNS firewall rules.	adc2.bb.wi.local and BB.WI.LOCAL resolved without the earlier delay.
Cross-forest DNS	Created conditional forwarders in each direction.	Each forest could resolve the other forest's namespace.
Recursion behaviour	Used no-recurse testing to separate authoritative-zone health from recursion delay; adjusted recursion for the internal lab phase.	Internal AD name resolution worked without root-hint timeout noise.

Conditional forwarders

ADC1: BB.WI.LOCAL -> 10.1.10.9
 ADC2: TT.WI.LOCAL -> 10.1.10.10

Conditional forwarders were the appropriate design because each forest remained authoritative for its own namespace while forwarding only queries for the partner forest to the correct domain DNS server. This avoided placing public resolvers on the domain-controller NICs and preserved Active Directory DNS behaviour.

Representative validation commands

```
ipconfig /all
ipconfig /flushdns
ipconfig /registerdns
Restart-Service DNS
nslookup adc1.tt.wi.local
nslookup adc2.bb.wi.local
nslookup 10.1.10.10
nslookup 10.1.10.9
dcdiag /test:dns
Test-NetConnection 10.1.10.9 -Port 53
```

DHCP role ADC1 also provided DHCP in the Windows lab. Windows clients used DHCP-assigned lab addresses during administration exercises, while DNS continued to point to the Active Directory DNS service.

5

Two-Way Active Directory Forest Trust

Implementation, authentication testing and security interpretation

Trust configuration

Setting	Implemented Value
Trust type	Forest Trust
Direction	Two-way
Transitivity	Transitive
Authentication mode used in lab	Forest-wide authentication
Forests	TT.WI.LOCAL <-> BB.WI.LOCAL
Primary objective	Cross-forest authentication after reliable DNS resolution was established

Validation evidence

- Local DNS resolution succeeded on ADC1 and ADC2.
- Cross-forest DNS resolution succeeded through the configured conditional forwarders.
- nltest /domain_trusts showed direct inbound and direct outbound trust relationships.
- BB\Administrator successfully authenticated on a TT-side Windows workstation.
- Domain-controller interactive logon remained subject to logon-right policy, demonstrating that a valid trust does not override authorization controls.

```
nltest /domain_trusts
nslookup adc2.bb.wi.local 10.1.10.10
nslookup adc1.tt.wi.local 10.1.10.9
```

Security considerations

- Forest trusts are high-impact security relationships because they expand where identities may be recognized.
- Forest-wide authentication is convenient for a controlled lab; selective authentication is a safer design to evaluate for separate organizations or higher-security production environments.
- Interactive logon to domain controllers should remain tightly restricted.
- Least privilege should be maintained across forests instead of relying on broad administrator membership.
- Trust creation and cross-domain authentication events should be monitored using Windows logging, SIEM or Windows Event Forwarding where available.
- DNS forwarders and recursion settings are operational dependencies and should be documented as part of the trust configuration.

Key lesson A trust can exist and validate while a logon attempt is still blocked by policy. Authentication proves identity; authorization decides whether that identity can perform the requested action.

6

File Services, NTFS Permissions & Recovery

SMB administration, folder redirection and Volume Shadow Copy recovery

SMB and NTFS administration

User access exercises included creating or administering server-hosted shares, assigning permissions and validating that intended users could access the required location without broadening access to unrelated users or folders. The work reinforced the distinction between share permissions, NTFS permissions and the effective access produced by both layers.

- Created/managed Windows SMB shares for user-access scenarios.
- Applied NTFS access-control entries using Windows ACL tooling and PowerShell/.NET security classes where required.
- Used folder redirection to place user Documents on a server-hosted path rather than relying only on local workstation storage.
- Validated access as the restricted user and confirmed policy/permission behaviour from the user context.

File recovery case study - Previous Versions / VSS

Stage	Evidence / Action
Incident	A file on a Windows network share was deleted and was not recoverable from the local Recycle Bin.
Finding	The shared folder exposed Previous Versions created from Volume Shadow Copy snapshots.
Recovery	Opened the appropriate point-in-time shadow copy and restored data using ROBOCOPY to a controlled recovery location.
Verification	Compared recovered content and used file-hash validation during the recovery workflow.
Operational lesson	Network-share deletion can bypass the user's local Recycle Bin; VSS provides point-in-time recovery when snapshots exist.

```
ROBOCOPY "\\DC\Shared\@GMT-<snapshot>\<folder>" "C:\Recovery\<target>" <files>
Get-FileHash <source-file> -Algorithm SHA256
Get-FileHash <restored-file> -Algorithm SHA256
```

Recovery principle A restore is not complete when a file merely reappears. The recovered data should be validated and the source snapshot/time should be recorded so the recovery is reproducible and auditable.

7

Least-Privilege Helpdesk Delegation

Separate GOODCORP.LOCAL administration challenge

Environment boundary This challenge was completed in GOODCORP.LOCAL and is included because it directly demonstrates enterprise Active Directory administration. It is not presented as part of the TT.WI.LOCAL / BB.WI.LOCAL forest topology.

Requirement

Create a Helpdesk administrative group that can create user accounts and reset user passwords only inside the UserAccounts OU, without receiving deletion rights, permission-management rights or domain-wide authority.

Item	Implementation
Domain	goodcorp.local
Administrative identity	GOODCORP\goodda (elevated PowerShell)
Delegated group	helpdesk - Global Security group
Members	helpdesk1 and helpdesk2
Delegation target	OU=UserAccounts,DC=goodcorp,DC=local
Allowed operations	Create User objects; Reset passwords for descendant User objects
Explicitly avoided	GenericAll, WriteDacl, WriteOwner, DeleteChild and domain-root delegation

Why object-specific ACEs were used

The built-in “create, delete and manage user accounts” shortcut was intentionally avoided because it bundled authority beyond the requirement. Instead, two precise access-control entries were applied to the UserAccounts OU.

Operation	Evaluated Against	AD Permission	Scope
Create a user	Parent UserAccounts OU	CreateChild restricted to the User object class	No inheritance beyond the required creation scope
Reset a password	Existing descendant User object	Reset Password extended right restricted to User objects	Applies only to descendant users in UserAccounts

Validator results

- helpdesk group found and membership matched the intended operators.
- Create User delegated on UserAccounts: PASS.
- Reset Password delegated on UserAccounts: PASS.
- Scope limited to UserAccounts with no domain-root grant: PASS.
- Non-helpdesk user remained outside the delegated group: PASS.

- No broad-rights warning detected: PASS.

Troubleshooting example: correct identity, wrong token

An initial New-ADGroup attempt returned Access is denied even though GOODCORP\goodda was the current identity. The PowerShell process did not have an elevated administrative token. The session was elevated and the token was explicitly checked before repeating the operation successfully.

```
whoami
$CurrentUser = [Security.Principal.WindowsPrincipal] [Security.Principal.WindowsIdentity]::GetCurrent()
$CurrentUser.IsInRole([Security.Principal.WindowsBuiltInRole]::Administrator)
Start-Process powershell.exe -Verb RunAs
```

Security lesson When an administrative requirement fails, verify identity, target and scope before adding more permissions. Broadening access to make a command succeed undermines least privilege.

8

PowerShell & Native Administration Tooling

Repeatable configuration, inspection and evidence collection

PowerShell patterns used

- ActiveDirectory module cmdlets for domain, user, group and OU operations.
- Reusable foreach loops for consistent user provisioning.
- SecureString input for temporary passwords so credentials were not hard-coded in scripts or documentation.
- ACL inspection and Set-Acl operations for object-specific delegation.
- Get-Service, Set-Service and Start-Service for Windows service-state remediation.
- Get-NetFirewallRule and Test-NetConnection for service/firewall/network validation.
- Tee-Object to display validator results while preserving an evidence file.
- Get-WinEvent, Event Viewer and Sysmon for Windows event visibility.

Representative Active Directory provisioning pattern

```
$DomainDN = (Get-ADDomain).DistinguishedName
$DNSRoot = (Get-ADDomain).DNSRoot
$UserNames = @("helpdesk1", "helpdesk2")
foreach ($UserName in $UserNames) {
    if (Get-ADUser -Filter "SamAccountName -eq '$UserName'") { continue }
    $Password = Read-Host "Temporary password for $UserName" -AsSecureString
    New-ADUser -Name $UserName -SamAccountName $UserName `
        -UserPrincipalName "$UserName@$DNSRoot" -AccountPassword $Password `
        -Enabled $true -Path "CN=Users,$DomainDN"
}
Add-ADGroupMember -Identity "helpdesk" -Members "helpdesk1", "helpdesk2"
```

Command families demonstrated

Administration Area	Representative Tools / Commands
Identity / AD	Get-ADDomain, Get-ADUser, New-ADUser, New-ADGroup, Add-ADGroupMember, Get-ADOrganizationalUnit
DNS / network	ipconfig, nslookup, dcdiag, netstat, ping, Test-NetConnection
Trust / domain	nltest /domain_trusts, gpupdate, gpresult, RSOP
Permissions / files	Get-Acl, Set-Acl, New-SmbShare, ROBOCOPY, Get-FileHash
Services / security	Get-Service, Set-Service, Start-Service, Get-NetFirewallRule, Get-WinEvent, Sysmon

Documentation practice Commands were treated as evidence, not as an end in themselves. The report records what each command was checking and how the result informed the next administration decision.

9

Operational Troubleshooting Case Studies

Service state, network access, permissions and endpoint diagnosis

OpenSSH service remediation on Windows

Stage	Finding / Action
Symptom	SSH connectivity to the Windows server failed.
Evidence	OpenSSH.Server was installed, but the sshd service was Stopped and configured for Manual startup. A related firewall rule also required validation/enabling.
Proven root cause	The SSH service was not running and would not start automatically.
Remediation	Set sshd to Automatic, started the service, and enabled the OpenSSH SSH Server firewall rule.
Validation	Confirmed TCP/22 listening and used Test-NetConnection from the client; the port test succeeded.

```
Get-Service sshd
Set-Service sshd -StartupType Automatic
Start-Service sshd
Get-NetFirewallRule -DisplayName "OpenSSH SSH Server (sshd)"
Test-NetConnection 10.1.6.11 -Port 22
```

Endpoint CPU / memory diagnostic workflow

A separate “computer running slow” investigation used PowerShell process measurements and session checks to compare CPU and memory consumption, inspect the affected user context and establish a baseline. No persistent high-CPU process was proven during the captured checks, so the final documentation avoided inventing a root cause.

- Measured process CPU and private-memory usage instead of relying only on subjective performance reports.
- Checked user sessions with quser and investigated disconnected-session state.
- Captured output to text for repeatable review and ticket evidence.
- Recorded the absence of a proven persistent offender instead of assigning an unsupported cause.

Troubleshooting discipline The documented root cause should be the condition proven to have caused the incident. Additional observations belong in investigation notes unless they were necessary to restore service.

10

Security Logging & Monitoring Extension

Sysmon and Windows telemetry as a bridge from administration to security operations

The Windows administration work was later extended with Sysmon and a Wazuh Windows agent. This does not change the role of the Active Directory lab; it adds security visibility to the systems being administered and provides richer process/event telemetry for later SOC investigations.

Sysmon administration

- Installed Sysmon64 on a Windows lab client using a configuration file.
- Verified the Sysmon service was running.
- Used the Microsoft-Windows-Sysmon/Operational event channel for enhanced process and system telemetry.
- Configured/used the Wazuh Windows agent in later SOC-lab work to collect Windows event telemetry.
- Connected systems-administration knowledge with security monitoring by understanding what “normal” Windows service, process and authentication behaviour looks like.

```
.\Sysmon64.exe -i <config-file.xml>
Get-Service Sysmon64
Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 20
```

Why this belongs in the administration portfolio

Secure administration includes observability. Domain, DNS, service and access-control changes should be visible through logs so administrators can verify expected behaviour and security teams can investigate deviations. Sysmon therefore complements - rather than replaces - the core Windows administration skills in this project.

Security bridge The portfolio story is not “administrator or SOC analyst.” The stronger progression is: administer Windows correctly -> understand normal identity/network/service behaviour -> instrument the systems -> investigate deviations with better context.

11

Project Outcomes & Lessons Learned

What the completed work demonstrates to a reviewer

Technical outcomes

Area	Outcome
Active Directory	Functional multi-forest lab with Windows clients, identity administration and policy testing.
DNS	Local and cross-forest name resolution restored and validated; conditional forwarders documented.
Forest trust	Two-way transitive forest trust created and verified; cross-domain workstation authentication demonstrated.
Group Policy	User restrictions, logon rights and folder-redirection controls applied and validated.
Delegation	Helpdesk create-user and reset-password rights implemented without broad deletion/ACL/ownership privileges.
File services	SMB/NTFS administration and VSS/Previous Versions recovery workflow practiced.
PowerShell	Configuration, automation, ACL manipulation, service troubleshooting and evidence collection used throughout.
Security visibility	Sysmon/Windows event telemetry added as a monitoring extension.

Key lessons

- DNS must be reliable before Active Directory trusts are configured or trusted as healthy.
- Reverse lookup and PTR records can materially affect how DNS server identity and troubleshooting results appear.
- IPv6 loopback precedence can cause confusing delays in small labs when DNS interfaces are not configured deliberately.
- dcdiag results must be interpreted in context; an isolated lab can legitimately behave differently from an internet-connected enterprise resolver.
- A trust relationship provides an authentication path, but access and logon rights remain governed by authorization policy.
- Least privilege is easier to defend when rights are expressed precisely at the correct AD object and OU scope.

- Administrative identity and process elevation are separate conditions; both should be verified during “Access is denied” troubleshooting.
- Recovery should be validated, not merely attempted.
- Clear documentation of the symptom, evidence, fix and validation creates a reusable operational runbook.

Skills demonstrated

Skill Domain	Evidence in Project
Windows Server	AD DS, domain controller administration, DNS, DHCP, file services
Identity & Access	Users, groups, OUs, delegation, password-reset rights, least privilege
Group Policy	User/device restrictions, logon rights, policy application and verification
Networking	DNS resolution, TCP/UDP service checks, conditional forwarding, trust dependencies
PowerShell	AD cmdlets, ACLs, services, firewall inspection, file recovery validation
Troubleshooting	Evidence-led diagnosis across DNS, policy, permissions, services and endpoints
Security Operations Foundation	Sysmon, Windows event logs, trust/security implications, monitoring mindset
Documentation	Implementation guides, ticket-quality notes, validation evidence and lessons learned

Portfolio takeaway The project demonstrates practical Windows administration with security-aware decision making: build the environment, apply controls, troubleshoot real failures, validate the result and document the reasoning.

A Command Reference Appendix

Representative commands used across the consolidated Windows administration work

A.1 DNS registration and service checks

```
net stop netlogon
net start netlogon
ipconfig /flushdns
ipconfig /registerdns
Restart-Service DNS
ipconfig /all
netstat -an | findstr ":53"
```

A.2 DNS / trust validation

```
nslookup TT.WI.LOCAL
nslookup BB.WI.LOCAL
nslookup adc1.tt.wi.local
nslookup adc2.bb.wi.local
nslookup 10.1.10.10
nslookup 10.1.10.9
dcdiag /test:dns
nltest /domain_trusts
```

A.3 Connectivity and firewall validation

```
ping 10.1.10.9
ping 10.1.10.10
Test-NetConnection 10.1.10.9 -Port 53
Get-NetFirewallRule | Where-Object {$_.DisplayName -like "*DNS*"} | Select DisplayName,Enabled
```

A.4 Group Policy

```
gpupdate /force
gpresult /r
rsop.msc
```

A.5 Active Directory / delegated administration

```
Import-Module ActiveDirectory
Get-ADDomain
Get-ADUser
New-ADUser
New-ADGroup
Add-ADGroupMember
Get-ADGroupMember
Get-ADOrganizationalUnit
Get-Acl "AD:\<DistinguishedName>"
Set-Acl -Path "AD:\<DistinguishedName>" -AclObject $ACL
```

A.6 Windows services / remote administration

```
Get-Service sshd
Set-Service sshd -StartupType Automatic
Start-Service sshd
Get-NetFirewallRule -DisplayName "OpenSSH SSH Server (sshd)"
Test-NetConnection <server-ip> -Port 22
```

A.7 File recovery and validation

```
ROBOCOPY <snapshot-source> <recovery-target> <files>  
Get-FileHash <file> -Algorithm SHA256
```

A.8 Event and security telemetry

```
Get-WinEvent -LogName "DNS Server" -MaxEvents 20  
Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 20  
.\\Sysmon64.exe -i <config-file.xml>
```

Evidence source documents consolidated

- Active Directory Forest Trust implementation documentation for TT.WI.LOCAL <-> BB.WI.LOCAL.
- GOODCORP Active Directory Helpdesk Delegation PowerShell documentation and validator output.
- Windows administration lab/ticket work covering Group Policy, file recovery, service troubleshooting and endpoint diagnostics.
- Windows/Sysmon telemetry work used later as part of the broader SOC lab.

Public portfolio note All addresses, domains, users and hostnames in this report refer to controlled lab environments. The document is intended to demonstrate technical process and does not disclose employer infrastructure.